

Oklahoma NEVI Web Portal

A statewide ArcGIS Experience Builder application that tracks EV charging infrastructure eligibility, funding awards, and buildout progress across Oklahoma's federally designated corridors

The Problem

Before this application, prospective grant applicants and transportation planners had no easy way to verify whether a candidate site met the federal NEVI program's core requirement, that a fast charging station sit within one drive-mile of an interstate exit and no more than 50 miles from the next compliant station along a designated corridor. Checking either of those rules by hand meant working with FHWA corridor designations, road network data, and a growing list of already-funded sites, all separately, for every candidate location under consideration. As funding rounds progressed and more sites came online, tracking which stretches of corridor still had a coverage gap and which award decisions had already been made became its own ongoing coordination problem.

My Approach

I built this as a public-facing **ArcGIS Experience Builder application**, so applicants, local utilities, and the public could check site eligibility and program status themselves rather than needing that information relayed by agency staff.

Eligibility zones computed the way a driver actually travels. Candidate site zones around each qualifying interchange are built from a network drive-distance analysis along public roads, not a circular buffer, so the one-mile eligibility boundary matches actual travel distance rather than straight-line distance from the exit.

Coverage gaps computed and shown directly, not left as a report. A separate gap layer runs the same network distance logic along the full length of each designated corridor and highlights any stretch exceeding 50 miles without a compliant station, turning "where does the program still need chargers" into something visible on the map instead of a separate spreadsheet exercise.

Award status as a first-class layer, not an afterthought. Round 1 award sites are shown with their construction status, so anyone, an applicant, a local official, or the public, can see at a glance which projects are funded, under construction, or already operational, without waiting on a status update from the agency.

Filtering built around how people actually use it. Rather than one large undifferentiated map, users can filter candidate sites and awards by corridor or by construction status, so someone interested in a specific interstate or a specific project phase isn't scrolling through the entire state to find it.

Data made available, not just visualized. Beyond the interactive map itself, the underlying candidate site layers can be downloaded directly from the application as shapefiles or geodatabases, and every layer's attribute table can be opened and sorted in place, so a user who needs to work with the data in their own GIS software doesn't have to ask for it separately.

Technical Highlights

- **Platform:** ArcGIS Experience Builder, built on ArcGIS Online hosted feature layers
- **Data sources:** ODOT GIS open data, FHWA alternative fuel corridor designations, US Department of Energy Alternative Fuels Data Center for existing chargers, ODOT NEVI program award records
- **Spatial analysis:** Network drive-distance polygons for interchange eligibility zones, network distance analysis along corridors to identify coverage gaps exceeding the 50-mile spacing requirement
- **Interactivity:** Corridor and status filtering, an in-app attribute data table, direct data download for candidate site layers, geocoded search, detailed feature pop-ups for sites, awards, and chargers
- **Scope:** Statewide coverage of Oklahoma's federally designated alternative fuel corridors across interstate and US/state routes, tracking a \$66 million federal allocation across multiple funding rounds

Why It Matters

A phased infrastructure funding program depends on applicants and planners being able to see, in real time, where eligible sites exist and where the remaining coverage gaps are, especially as the program moves from interstate corridors in its first round to state and US routes in later rounds. Putting the actual eligibility geometry, award status, and downloadable data in one public application meant Round 1's twelve awarded sites and \$8.2 million in funding could be evaluated against a shared, visible standard, and gives Round 2 planning the same clear picture of what's left to reach full corridor buildout.